

WattCast Rule Engine — Design Document

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Overview

WattCast's rule engine is the core of its value proposition. It extends Solar Assistant's basic automation model with predictive, future-aware conditions that SA will never have — loadshedding schedule awareness, PV forecasting, weather data, and smart switch control.

Design Principles

1. **Installer has full control** — all rules are configurable, including the core logic. WattCast ships with sensible defaults but nothing is locked.
 2. **Homeowner can view, not edit** — rules are visible to the homeowner for transparency, but only the installer (or WattCast admin) can modify them.
 3. **First rule wins** — when multiple rules target the same setting simultaneously, the highest-priority rule (lowest index / topmost in the list) takes effect. Priority ordering is a future discussion.
 4. **Hard safety limits always apply** — regardless of what a rule says, WattCast will never write values outside safe bounds (e.g. never below `output_shutdown_capacity`).
 5. **Default rules work out of the box** — a new install with default rules should immediately provide value without any installer configuration.
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Rule Structure

Inspired by Solar Assistant's automation model, extended with predictive conditions.

Each rule has:

- **Name** — human-readable label (e.g. "Pre-loadshedding charge boost")
- **When** — one or more trigger conditions (AND logic between conditions)
- **Set** — what to change and what value to set it to
- **Priority** — order in which conflicting rules are resolved (lower = higher priority)
- **Enabled/Disabled** toggle

Example rule (natural language)

When loadshedding starts in less than 3 hours
And when battery SoC < 80%
And when PV forecast remaining today > 1 kWh
Set Capacity point 1-6 = 80%

Condition Palette ("Based on which condition(s)?")

Current state conditions (SA has these — WattCast inherits)

CONDITION	TYPE	EXAMPLE
Time of day	Time range	00:00 to 06:00
Day of week	Multi-select	Mon, Tue, Wed
Month of year	Multi-select	Jun, Jul, Aug (winter)
Battery state of charge	Range %	0% to 25%
Battery voltage	Range V	44V to 48V
Grid voltage	Range V	0V to 180V (outage detection)
Grid frequency	Range Hz	49.5 to 50.5
PV energy progress today	Range kWh	0 to 5 kWh
PV energy remaining today	Range kWh	< 2 kWh

Predictive conditions (WattCast-exclusive — SA will never have these)

CONDITION	TYPE	EXAMPLE
Loadshedding starts in	Duration	< 3 hours
Loadshedding ends in	Duration	< 1 hour
Loadshedding duration	Duration	> 2 hours
Loadshedding stage	Range	>= 2
Loadshedding active	Boolean	Yes / No
Next loadshedding slot today	Boolean	Yes / No
PV forecast tomorrow	Range kWh	< 5 kWh (cloudy day)
PV forecast today remaining	Range kWh	> 3 kWh
Current irradiance	Range W/m ²	> 400 W/m ²
Last 30min avg irradiance	Range W/m ²	< 100 W/m ² (cloud cover)
Weather condition	Select	Cloudy / Clear / Rain
Season	Select	Summer / Winter / Shoulder

Compound conditions

Multiple conditions can be combined with AND logic (same as SA's "And when" model). OR logic is achieved by creating multiple rules targeting the same setting.

Action Palette ("Which setting do you want to control?")

Inverter settings — Safe (green)

Standard operational settings, safe to automate:

- Capacity point 1–6 (battery discharge floor per TOU slot)
- Charge point 1–6 (charge target per TOU slot)
- Grid charge point 1–6
- Start battery discharge capacity / voltage
- Stop battery discharge capacity / voltage
- Start grid charge capacity / voltage
- Use timer (on/off)
- Work mode

- Energy pattern
- Grid charge (on/off)
- Solar export when battery full
- Time point 1–6
- Program point 1–6
- Sell point 1–6
- Power point 1–6

Inverter settings — Caution (⚠ warning shown)

Can cause unexpected behaviour if set incorrectly:

- Max charge current
- Max discharge current
- Max grid charge current
- Max sell power
- Max solar power
- Output shutdown capacity
- Output shutdown voltage
- Grid voltage high/low
- Grid frequency high/low
- Grid peak shaving / Grid peak shaving power
- Grid trickle feed
- Battery float charge voltage
- Battery absorption charge voltage
- Battery equalization charge voltage

Inverter settings — Restricted (🔒 installer-only or hidden)

Protocol/hardware level — not exposed to rule engine by default:

- Battery type
- Battery operation
- Lithium protocol
- Auxiliary port

- Remote switch
- Generator settings (unless site has a generator)
- Battery capacity (set once at install, not automated)

Smart switches (WiFi-enabled) — WattCast extension

WattCast extends the action palette beyond inverter settings to include WiFi smart switches.

This is a key differentiator — SA cannot control external devices.

Target device types:

- Geyser / water heater (high priority — major load, 2–3kW)
- Pool pump
- EV charger
- Air conditioner
- Irrigation system
- Any other WiFi smart plug / switch

Supported protocols — Phase 1 (webhooks + MQTT):

- **Webhooks** — device calls a WattCast HTTP endpoint on state change; WattCast calls device HTTP endpoint to control it. No special protocol needed.
- **MQTT** — device publishes state to MQTT broker; WattCast publishes to device control topic. Since WattCast already has MQTT running, this is nearly free to add.

Devices covered by Phase 1:

- **Shelly** — webhooks + MQTT natively, no cloud dependency, local-first 
- **Sonoff with Tasmota** — MQTT natively 
- **Sonoff stock firmware** — webhooks 
- **Many Tuya devices** — MQTT via local API or Tasmota flash 

Phase 2 (future):

- Matter / Thread
- Home Assistant integration

- Tuya cloud API (for devices that can't be flashed)

Example smart switch rules:

- "When loadshedding active → turn off geyser"
 - "When PV power > 2kW AND battery SoC > 80% → turn on geyser"
 - "When grid outage detected → turn off pool pump"
 - "When PV forecast tomorrow < 3kWh → turn on geyser now while sun available"
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UI Design Considerations

Installer view (full access)

- Add / edit / delete / reorder rules
- Enable / disable individual rules
- Set rule priority (drag to reorder, or explicit priority number)
- See rule execution history (when did it last fire, what did it set)
- Configure site parameters (battery size, load profile, loadshedding area)

Homeowner view (read-only)

- See active rules and their current status
- See what WattCast last did and why ("Raised capacity point to 80% — loadshedding in 2h")
- See current inverter state and smart switch states
- Cannot edit rules (can request changes via installer)

Rule conflict display

- When two rules conflict, show which rule "won" and why
 - Highlight disabled rules that would have fired
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SA Automation Model vs WattCast Rule Engine

FEATURE	SOLAR ASSISTANT	WATTCAST
Time of day conditions	✓	✓
Battery SoC conditions	✓	✓
Grid/PV conditions	✓	✓
Loadshedding awareness	✗	✓
PV forecast conditions	✗	✓
Weather conditions	✗	✓
Smart switch control	✗	✓
Compound AND conditions	✓	✓
Compound OR conditions	✗ (workaround)	✓ (multiple rules)
Homeowner visibility	✗	✓ (read-only)
Multi-site management	✗	✓ (future)
Rule priority/conflict resolution	✗	✓

Rule Set Portability

A key installer workflow feature — build once, deploy many times.

Export

- Export a site's full rule set as a JSON/YAML file
- Includes all rules, priorities, parameters, and device mappings
- Installer can save named templates (e.g. "Small household 5kWh no generator")

Import

- Import a rule set file onto a new site
- WattCast prompts to map template device names to actual local devices (e.g. "Geyser" → select from discovered MQTT/webhook switches)
- Existing rules can be kept, replaced, or merged

Reset to defaults

- "Wipe and reinstate default rule set" option — removes all custom rules,

reinstates the factory defaults

- Requires installer-level confirmation
- Useful when a customer's rules are in a broken/conflicting state

Template library

- WattCast ships with a built-in template library
 - Installer can contribute/save their own templates
 - Future: shared template marketplace across installers
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Default Rule Set

Priority-ordered rules that ship with every WattCast install. All can be edited or disabled. Loadshedding rules are included but low priority since loadshedding has been suspended since

March 2025 — they will activate automatically when loadshedding returns.

Rule 1 — Solar surplus geyser (Priority 1 — highest impact daily)

When PV power > load power + 500W
And when Battery SoC > 80%
And when Time of day 08:00–16:00
Set Geyser switch = ON

Rationale: Biggest daily saving for most households. A 3kW geyser running 2hrs on free solar instead of grid = meaningful monthly bill reduction. 500W buffer ensures battery still charges while geyser runs.

Rule 2 — Geyser off at night / grid

When PV power < 200W
Set Geyser switch = OFF

Rationale: Ensure geyser never runs on battery or grid unnecessarily.

Rule 3 — Grid outage load shedding (unplanned outages)

When *Grid voltage < 180V*
Set *Pool pump = OFF*
Set *Geyser = OFF*
Set *EV charger = OFF*

Rationale: Unplanned outages still happen regardless of loadshedding schedule. Immediately shed non-essential loads to extend battery life.

Rule 4 — Overnight battery protection

When *Time of day 22:00–06:00*
And when *Battery SoC < 30%*
Set *Capacity point 1–6 = 25%*

Rationale: Prevent battery from draining to shutdown overnight. 25% floor gives enough buffer for morning until solar kicks in.

Rule 5 — Winter morning grid top-up

When *Month of year Jun, Jul, Aug*
And when *Time of day 06:00–08:00*
And when *Battery SoC < 50%*
And when *PV forecast today < 5 kWh*
Set *Grid charge = ON*
Set *Capacity point 1 = 60%*

Rationale: On cloudy winter mornings, charge from grid during off-peak before morning load spike. Only fires when solar forecast suggests poor generation.

Rule 6 — Solar peak optimiser

When *Time of day 10:00–14:00*
And when *PV power > 3000W*
Set *Capacity point 1–6 = 20%*

Rationale: During peak solar hours, lower the discharge floor so the battery absorbs as much solar as possible rather than exporting to grid at low value.

Rule 7 — Pre-loadshedding charge boost (inactive until loadshedding returns)

When Loadshedding starts in < 3 hours
And when Battery SoC < 80%
Set Capacity point 1–6 = 80%
Set Grid charge = ON

Rationale: Ensure battery is sufficiently charged before a loadshedding slot.
Target SoC calculated from slot duration × estimated load / battery capacity.
Currently dormant — will activate automatically when loadshedding resumes.

Rule 8 — Post-loadshedding recovery (inactive until loadshedding returns)

When Loadshedding active = No (just ended)
And when Time of day 06:00–18:00
Set Grid charge = OFF
Set Capacity point 1–6 = 60%

Rationale: After loadshedding ends, restore normal settings and let solar recover the battery naturally rather than continuing to grid charge.

Open Questions

1. ~~Smart switch protocol priority~~ **DECIDED**: Webhooks + MQTT first. Matter/Thread in Phase 2.
2. **Rule priority mechanism** — drag-to-reorder vs explicit number vs timestamp?
3. **OR logic** — is "create two rules" sufficient or do we need explicit OR operators?
4. ~~Default rule set~~ **DECIDED**: See Default Rule Set section above.
5. **Rule templates** — installer selects from pre-built templates and customises?
6. **Homeowner override** — can homeowner temporarily pause a rule? (e.g. "pause for today")
7. **Rule scheduling** — can rules be active only on certain days/dates (e.g. seasonal rules)?
8. **Geyser SoC target calculation** — how does WattCast calculate the right capacity point for loadshedding? Formula: $(\text{slot_duration_hrs} \times \text{avg_load_W}) / \text{battery_capacity_Wh} \times 100 + \text{safety_margin}$
9. **Webhook security** — how do we authenticate inbound webhooks from switches?

Next Steps

1. Design the default rule set for a typical install
2. Investigate Tuya local API vs Shelly for smart switch integration
3. Build the rule engine data model (Python dataclasses or similar)
4. Build a simple web UI for rule management (tech stack TBD)
5. Wire up first real rule: "Pre-loadshedding capacity boost"